LETHE · WHY NOT JUST ENCRYPT IT?

Cryptography hides who you are. A local model hides what you say.

WHO YOU ARE

wraps every request you send today

- **account** your email
- **card** your billing name
- **IP address** your city
- **device** its fingerprint
- **history** every past chat, linked



today: joined into one profile, per lab

CRYPTOGRAPHY hides who you are, completely: six ideas from crypto-currency — ✓ SOLVED

FROM E-CASH · CHAUM, 1982
Blind signatures
You buy credentials in bulk. The issuer signs them blind, so it can't tell which one you spend.
HIDES **who paid for what**

FROM ZCASH
Zero-knowledge
Prove "I paid" or "I'm a member" without revealing which member.
HIDES **which member asks**

FROM BITCOIN COINJOIN
Mixing
The pool: every member's requests look identical, padded to fixed sizes.
HIDES **which request is yours**

FROM MONERO
One-time keys
A fresh credential and key for every request. Nothing on the wire ever repeats.
HIDES **links between chats**

FROM BLOCKCHAINS
Public ledger
Everyone checks they got the same issuer key, so no one can be quietly tagged.
HIDES **per-user tagging**

FROM MULTISIG NEW
Threshold keys
Cutting off an abuser takes 2 of 3 independent keyholders. Nobody acts alone.
HIDES **no one acts alone**

2-OF-3 CUT-OFF NEW

proposed · from multisig wallets



a. a lab flags misuse
b. 2 of 3 keyholders sign
c. that subscription is cut off
d. the cut-off is publicly logged

Unlocks only which subscription to stop. **Never** what was asked, or who you are, to the lab.

WHAT YOU SAY

the words inside the request

Ethan, 9, has ADHD. Should he switch from Adderall? We're on **Kaiser** in **Sacramento**.

Names, insurer, city, condition, even your writing style point back to one person.

50–75%
of users still re-identified in at least one chat after their text was rewritten
UMass study, Sep 2026

THE LIMIT the model must read your words to answer — LIMIT OF MATH, TODAY

METHOD	MODEL READS	LAB MUST	SLOWDOWN vs. A NORMAL REQUEST
Send as is how every app works today	✗ everything, plus who you are	nothing	1×
Enclave (TEE) sealed hardware	✗ everything; only the server operator is blind	run the model in an enclave	≈1×
Multi-party computation split the maths across parties	✓ nothing	✗ hand out its model weights	10^3–10^4×
Homomorphic encryption compute on encrypted data	✓ nothing	✗ rebuild how it serves models	10^4–10^6×
Lethe: on-device model drops identifying details	✓ the question, not who asked	✓ nothing	≈1× on your device

1× 10 10^2 10^3 10^4 10^5 10^6×

log scale: each step is ten times slower

WHAT THE LAB GETS

one orphan question, at full capability



GPT · Claude · Gemini

My **child**, 9, has ADHD. Should they switch from Adderall? We're on **an HMO** in **California**.

✓ a valid, single-use credential
✗ account · card · IP · past chats

The answer comes back and your device puts **Ethan** back in.

No profile · full frontier model

Slowdowns are orders of magnitude from published research prototypes on BERT- and GPT-2-size models; none has run a frontier model. Re-identification: UMass study (Sep 2026). The 2-of-3 cut-off is a proposed design.

Lethe · Pranjal Joshi · Prashil Bulbule · Vaibhav · October 2026